KENDRIYA VIDYALAYA, EMBASSY OF INDIA, KATHMANDU, NEPAL

PRACTICE PAPER 02 (2025-26)

CHAPTER 02 THE INVISIBLE LIVING WORLD: BEYOND OUR NAKED EYE

(ANSWERS)

SUBJECT: SCIENCE **MAX. MARKS : 40**

CLASS : VIII **DURATION : 1½ hrs**

General Instructions:

(i). **All** questions are compulsory.

(ii). This question paper contains 2**0** questions divided into five Sections A, B, C, D and E.

(iii). **Section A** comprises of **10 MCQs** of **1 mark** each. **Section B** comprises of 4 questions of **2 marks** each. **Section C** comprises of 3 questions of **3 marks** each. **Section D** comprises of 1 question of **5 marks** each and **Section E** comprises of 2 Case Study Based Questions of **4 marks** each.

(iv). There is no overall choice.

(v). **Use of Calculators is not permitted**

SECTION – A

Questions 1 to 10 carry 1 mark each.

1. You find a powdery, white growth on a forgotten lemon. Based on the chapter, which microorganism is most likely responsible for this growth?
(a) Bacteria (b) Protozoa (c) Algae (d) Fungi
Ans. (d) Fungi
A powdery or cotton-like growth on food items like lemons and oranges is caused by microbes, specifically mentioning fungi and bacteria as types of microorganisms that act on such waste.

2. According to the chapter, what is the primary function of the cytoplasm in a cell?.
(a) It regulates all cell activities. (b) It provides rigidity and strength.
(c) It allows the entry and exit of materials. (d) Most life processes take place within it.
Ans. (d) Most life processes take place within it.
The text states, "Most of the life processes take place within the cytoplasm".

3. Which of the following is an example of a unicellular fungus?
(a) Bread mould (b) Mushroom (c) Yeast (d) Algae
Ans. (c) Yeast
Yeast is a unicellular fungus while mould is a multicellular fungus.

4. A botanist discovers a new plant species. When observing its cells under a microscope, which component would confirm that it is a plant cell and not an animal cell?
(a) Cell membrane (b) Cytoplasm (c) Nucleus (d) Cell wall
Ans. (d) Cell wall
Some cells, "like onion peel cells, have an extra outer layer called the cell wall," which is a key difference between plant and animal cells.

5. A group of cells work together to perform a specific function, such as protecting the inner lining of the cheek. According to the levels of organization, what do these cells form?
(a) An organ (b) A tissue (c) An organ system (d) An organism
Ans. (b) A tissue
A tissue as "a group of similar cells". The inner cheek cells, being thin and flat, form a protective lining, which is an example of a tissue.

6. Which scientist is known as the "Father of Microbiology" for being the first to clearly see and describe tiny living things like bacteria?

(a) Robert Hooke (b) Antonie van Leeuwenhoek
(c) Ananda Mohan Chakrabarty (d) Louis Pasteur
Ans. (b) Antonie van Leeuwenhoek
The text states that Antonie van Leeuwenhoek is known as the "Father of Microbiology" because he was the first person to clearly see and describe tiny living things like bacteria and blood cells.

7. What is a key characteristic that distinguishes bacteria from other microorganisms like yeast, protozoa, and algae?
(a) They are unicellular. (b) They lack a cell wall.
(c) They lack a well-defined nucleus and nuclear membrane. (d) They are found everywhere.
Ans. (c) They lack a well-defined nucleus and nuclear membrane.
Bacteria do not have a well-defined nucleus and nuclear membrane and instead have a nucleoid. This feature distinguishes them from cells of yeast, protozoa, algae, fungi, plants, and animals.

8. Which of the following is a non-cellular material present in an ostrich egg that helps in its continued development?
(a) The yolk (b) The shell and white liquid
(c) The nucleus (d) The cytoplasm
Ans. (b) The shell and white liquid
The egg contains "extra non-cellular material: a shell for protection and a white liquid that nourishes the cell during its continued development".

In the following questions 9 and 10, a statement of assertion (A) is followed by a statement of Reason (R). Choose the correct answer out of the following choices.
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

9. **Assertion (A):** The use of glycerin is important when preparing a slide of onion peel.
Reason (R): Glycerin prevents the cells from drying out and improves the clarity for better visualization of the cells.
Ans. (a) Both A and R are true, and R is the correct explanation of A.
The reason R directly explains the purpose and benefit of using glycerin (A) in the experiment.

10. **Assertion (A):** The elongated and branched structure of a nerve cell is a key feature that helps it perform its function.
Reason (R): Cells are found in different shapes and sizes because their unique structure helps them carry out specific functions.
Ans. (a) Both A and R are true, and R is the correct explanation of A.
The reason R provides a general principle (structure relates to function) that directly explains why the nerve cell's specific shape (A) is important for its function.

SECTION – B

Questions 11 to 14 carry 2 marks each.

11. A student notices that the letters on a book appear larger when viewed through a flask filled with water. Explain why this happens, based on the chapter.
Ans. This happens because the round-bottom flask filled with water acts as a magnifying glass. The curved surface of the flask and the water inside bend light in a way that makes the letters on the book appear larger than they are, allowing them to be seen more clearly.

12. Explain the function of plastids in plant cells and name the specific type of plastid responsible for the green color of plants.

Ans. Plastids are tiny rod-shaped structures found in all parts of a plant. In non-green parts of a plant, they help in the storage of substances. The specific type of plastid responsible for the green color of plants is the chloroplast, which contains chlorophyll and helps in the process of photosynthesis.

13. What is a foldscope, and what is its main advantage over a traditional laboratory microscope?
Ans. A foldscope is a low-cost, foldable paper microscope. Its main advantage is that it makes the microscopic world accessible to many people because it is affordable and portable.

14. Why are microorganisms, like bacteria and fungi, unable to grow on food items that have a high concentration of salt or sugar, such as pickles and murabbas?
Ans. Microorganisms cannot grow on foods with a high concentration of salt or sugar because these substances act as preservatives. A high concentration of salt or sugar creates an environment that is not conducive for microbial growth. This is because it draws out water from the cells of the microorganisms, essentially dehydrating them and preventing their multiplication.

SECTION – C

Questions 15 to 17 carry 3 marks each.

15. Describe the experiment mentioned in the chapter to observe human cheek cells. What are the key steps involved and what does a student observe?
Ans. To observe human cheek cells, a student must first rinse their mouth with clean water. Then, using the blunt end of a clean toothpick, they gently scrape the inside of their cheek to collect material. This scraped material is placed in a drop of water on a clean glass slide and spread evenly. A drop of methylene blue stain is added to the material to improve visibility and contrast. After a minute, a drop of glycerin is added to prevent the cells from drying. Finally, a coverslip is carefully placed over the material, and the slide is observed under a microscope. Under the microscope, a student would observe polygon-shaped structures, which are the cheek cells.

16. Explain the significance of the invention of the microscope. How did the contributions of both Robert Hooke and Antonie van Leeuwenhoek advance our understanding of the invisible living world?
Ans. The invention of the microscope was a major scientific discovery as it opened up a hidden world of tiny living creatures that were previously invisible to the naked eye. Robert Hooke's microscope, which could magnify objects 200 to 300 times, allowed him to see and name the 'cell' for the first time by observing a thin slice of cork. Around the same time, Antonie van Leeuwenhoek improved lenses to build more useful microscopes, which allowed him to be the first person to clearly see and describe tiny living organisms like bacteria and blood cells. His work earned him the title of the "Father of Microbiology". Together, their work laid the foundation for the field of microbiology and our understanding of the fundamental unit of life.

17. Explain how microorganisms help to recycle nutrients in nature. Give an example of this process.
Ans. Microorganisms play a vital role in recycling nutrients in nature. They act as decomposers, breaking down dead plant and animal waste into simpler, nutrient-rich substances. These nutrients are then returned to the soil, where they help new plants grow better. A specific example of this is the process of manure formation. Some microorganisms, like fungi and bacteria in the soil, act on plant waste (e.g., fruit and vegetable peels) and slowly break it down into natural manure. This helps to increase the fertility of the soil and maintains the nutrient balance in the environment.

SECTION – D

Questions 18 carry 5 marks each.

18. In the chapter a cell is compared to a brick and an organism to a complete building. Using this analogy and the levels of organization described, elaborate on how different cells in an organism work together to perform complex functions, using the digestive system as a specific example.

Ans. The analogy of a cell as a brick and an organism as a building helps to understand the levels of organization in a living being. Just as a group of bricks forms a wall, a group of similar cells forms a tissue. Different tissues then organize to form an organ, and several organs working together form an organ system, which finally makes up a complete organism.

The digestive system is an excellent example of this organization. It is an organ system responsible for digestion. Different parts of the digestive system are made up of different types of cells, all working together for a common goal.

Cells to Tissue: A group of muscle cells in the food pipe, for instance, forms a tissue. These cells are thin, flexible, and spindle-shaped, allowing them to contract and relax in a wave-like manner to push food down to the stomach.

Tissue to Organ: The stomach itself is an organ composed of different tissues. One type of muscle tissue in the stomach wall helps to churn the food. Another tissue, composed of different types of cells in the inner lining, produces digestive juices and acid to break down the food chemically.

Organ to Organ System: The stomach, along with other organs like the food pipe and intestines, works together to form the digestive system. All these cells, tissues, and organs cooperate to ensure digestion is possible, highlighting how simple building blocks (cells) come together to perform a complex function.

SECTION – E (Case Study Based Questions)

Questions 19 to 20 carry 4 marks each.

19. Case Study-1: The Soil's Secret Helpers

A school science club was running an experiment to learn about composting. They took two identical containers. Container A was filled with garden soil, and Container B was filled with garden soil and a mix of kitchen waste like fruit and vegetable peels. They covered both containers and left them in a corner of the school garden. After a few weeks, they noticed that the contents of Container B had turned into a dark-coloured, nutrient-rich material, while the contents of Container A remained unchanged. The science teacher explained that tiny living beings in the soil were responsible for this transformation. She also mentioned that some gardeners in the neighbourhood use a similar technique.

Based on the passage and the chapter, answer the following questions:

(a) What is the dark-coloured material in Container B called, and what is its significance?

(b) What type of microorganisms are responsible for the change in Container B?

(c) Why did the contents of Container A remain unchanged?

(d) How does this process help the environment?

Ans. (a) The dark-coloured material in Container B is called manure. Its significance is that it is rich in nutrients and helps to increase the fertility of the soil, making it beneficial for plant growth.

(b) The microorganisms responsible for the change are fungi and bacteria. They act on the plant waste and slowly break it down into simpler substances.

(c) The contents of Container A remained unchanged because it contained only soil and no biodegradable organic waste like fruit and vegetable peels. The microorganisms in the soil had nothing to decompose, so no manure was formed.
(d) This process helps the environment by recycling waste and returning important nutrients to nature. It reduces the amount of waste that would otherwise go to landfills and provides a natural way to enrich the soil without using chemical fertilizers.

20. Case Study-2: A Scientist's Innovative Solution

Dr. Ananda, a brilliant scientist, was deeply concerned about the environmental damage caused by oil spills in the oceans. He knew that conventional methods were often slow and ineffective. He decided to find an alternative solution. After years of research, in 1971, he successfully engineered a special bacterium that had the unique ability to break down oil. This discovery was so significant that it was granted a patent in 1980, ensuring that no one could copy his invention without permission. His work demonstrated a novel way to use microorganisms to tackle major environmental problems.

Based on the passage and the chapter, answer the following questions:
(a) What was the main environmental problem Dr. Ananda aimed to solve?
(b) What special ability did the bacterium he developed have?
(c) What is a 'patent' and why was it important for Dr. Ananda's discovery?
(d) How did Dr. Ananda's work expand our understanding of microorganisms?
Ans. (a) The main environmental problem Dr. Ananda aimed to solve was the pollution caused by oil spills in the ocean.
(b) The special ability of the bacterium he developed was that it could break down oil spills.
(c) A patent is a legal copyright given to a person so that no one else can copy, use, or sell their invention without permission. It was important for Dr. Ananda's discovery because it protected his intellectual property and ensured that his unique solution could not be exploited without his consent.
(d) Dr. Ananda's work showed that microorganisms are not just for basic life processes or decomposition but can also be used to solve complex environmental problems like pollution. It expanded our understanding of the potential and versatility of microbes beyond their traditionally known roles.